Cooperative Learning of Shared Latent Dynamics for Distributed Online Optimization: Supplementary material

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This document contains the proofs of the results of the paper *Cooperative Learning of Shared Latent Dynamics for Distributed Online Optimization*, referred to as the main paper. Section 1 proves Lemma 1; Section 2 states and proves the concentration bound on the sample moments (Lemma S1) on which Theorem 2 rests; Sections 3–5 prove Theorem 2, Lemma 3, and Theorem 4; and Appendix A collects the auxiliary results used along the way. The notation, the assumptions, and the results are those of the main paper and are not restated: references without the prefix S, such as (2), Assumption 2, Lemma 1, or Algorithm 1, point to the main paper, whereas the equations and the lemmas that appear only here are numbered S1, S2, ... Throughout, $\|\cdot\|$ is the Euclidean norm of a vector and the spectral norm of a matrix, $\|\cdot\|_F$ is the Frobenius norm, $\sigma_{\min}(\cdot)$, $\lambda_{\min}(\cdot)$, and $\lambda_{\max}(\cdot)$ are the smallest singular value and the extreme eigenvalues, $\rho(\cdot)$ is the spectral radius, $(x)_+ := \max\{x, 0\}$, e_a is the a-th canonical basis vector, $\mathrm{col}(v_1, \dots, v_m)$ stacks the vectors $v_1, \dots, v_m$, and $J := \mathbf{1}_m\mathbf{1}_m^\mathsf{T}/m$, so that $\sigma = \|W - J\|$. Every proof is organized in numbered steps, and later steps refer to the displayed equations of earlier ones.

1. Proof of Lemma 1 (reconstruction error)

Lemma 1 asserts, for $k \geq k_0$, the excitation bound $\bar{C}_{i,t}^\mathsf{T}\bar{R}_i^{-1}\bar{C}_{i,t} \succeq \alpha_{k,i}I_p$, the covariance bound $\Sigma_{\eta,i}(t) \preceq \alpha_{k,i}^{-1}I_p$, and the error bound (14). Its proof also establishes the properties of the noise η_i and the bias β_i of (6) that the main paper states after (6) and that the later proofs use (Step 2), and a bound on the second moments of the window estimates (Step 5). We write

$$b_{k,i} := \Gamma_i C_\rho \chi_i \Big(\frac{k}{\sqrt{3}} \|A - I\| B_{\theta,i} + \sqrt{\frac{k}{2} \operatorname{tr} Q_i} \Big) \tag{S1}$$

for the bias part of the bound $\epsilon_i^{\mathrm{w}} = \sqrt{p/\alpha_{k,i}} + b_{k,i}$ of (14).

Proof *Step 0 (moments of the parameters).* By Assumption 2, $\rho(A) < 1$, so Gelfand's formula yields $C_\rho \geq 1$ and $\rho_A \in (\rho(A), 1)$ with $\|A^j\| \leq C_\rho \rho_A^j$ for all $j \geq 0$, and the series $\Sigma_{\theta,i} = \sum_{l\geq 0} A^l Q_i (A^l)^\mathsf{T}$ converges; it is the unique solution of the Lyapunov equation $\Sigma_{\theta,i} = A\Sigma_{\theta,i}A^\mathsf{T} + Q_i$, and it is positive definite because $(A, Q_i^{1/2})$ is controllable. Iterating (2) from $\theta_i(0) \sim \mathcal{N}(0, P_{i,0})$, with the process noises zero mean and independent of each other and of $\theta_i(0)$, the second moment $P_i(t) = \mathbb{E}[\theta_i(t)\theta_i(t)^\mathsf{T}]$ satisfies

$$P_i(t) = A^t P_{i,0} (A^t)^\mathsf{T} + \sum_{l=0}^{t-1} A^l Q_i (A^l)^\mathsf{T}, \qquad P_i(t) - \Sigma_{\theta,i} = A^t \big(P_{i,0} - \Sigma_{\theta,i}\big)(A^t)^\mathsf{T}, \tag{S2}$$

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so that $B_{\theta,i} = \sup_{t\geq 0}(\operatorname{tr} P_i(t))^{1/2} \leq (C_\rho^2 \operatorname{tr} P_{i,0} + \operatorname{tr}\Sigma_{\theta,i})^{1/2}$ is finite, and $(\mathbb{E}\|\theta_i(t)\|^2)^{1/2} = (\operatorname{tr} P_i(t))^{1/2} \leq B_{\theta,i}$ for every t.

Now fix $i \in \mathcal{V}$ and $t \in \{0,\dots,N-k\}$, and write $\theta_i(t+j) = \theta_i(t)+\Delta_{i,t}(j)$ for $j = 0,\dots,k-1$, where, by iterating (2),

$$\Delta_{i,t}(j) := \theta_i(t+j) - \theta_i(t) = (A^j - I)\theta_i(t) + \sum_{l=0}^{j-1} A^{j-1-l} w_i(t+l), \qquad \Delta_{i,t}(0) = 0. \tag{S3}$$

Stacking the k measurements $\tilde{y}_i(t+j) = C_{i,t+j}\theta_i(t+j) + w_{m,i}(t+j)$ of the window gives the window model

$$\bar{y}_{i,t} = \bar{C}_{i,t}\theta_i(t) + \bar{b}_{i,t} + \bar{w}_{m,i,t}, \qquad \bar{b}_{i,t} = \operatorname{col}\big(C_{i,t}\Delta_{i,t}(0),\dots,C_{i,t+k-1}\Delta_{i,t}(k-1)\big), \tag{S4}$$

where $\bar{w}_{m,i,t} = \operatorname{col}(w_{m,i}(t),\dots,w_{m,i}(t+k-1)) \sim \mathcal{N}(0,\bar{R}_i)$ with $\bar{R}_i = I_k \otimes R_i$, because the measurement noises of different times are independent.

Step 1 (excitation). Since $k \geq k_0$, the window $\{t,\dots,t+k-1\}$ contains the $\lfloor k/k_0 \rfloor$ disjoint sub-windows $\{t+qk_0,\dots,t+(q+1)k_0-1\}$, $q = 0,\dots,\lfloor k/k_0\rfloor - 1$. Each of them starts at an index $t+qk_0 \leq t+k-k_0 \leq N-k_0$, so Assumption 3 applies to each of them. Since $\bar{R}_i^{-1} = I_k \otimes R_i^{-1} \succeq \lambda_{\max}(R_i)^{-1} I_{kn}$ and the matrices $C_{i,s}^{\mathsf{T}} C_{i,s}$ are positive semidefinite,

$$\begin{aligned}\bar{C}_{i,t}^{\mathsf{T}}\bar{R}_i^{-1}\bar{C}_{i,t} &\succeq \frac{1}{\lambda_{\max}(R_i)}\sum_{j=0}^{k-1} C_{i,t+j}^{\mathsf{T}} C_{i,t+j} \succeq \frac{1}{\lambda_{\max}(R_i)}\sum_{q=0}^{\lfloor k/k_0\rfloor -1}\sum_{j=0}^{k_0-1} C_{i,t+qk_0+j}^{\mathsf{T}} C_{i,t+qk_0+j} \\ &\succeq \frac{\lfloor k/k_0\rfloor \alpha_0}{\lambda_{\max}(R_i)} I_p = \alpha_{k,i} I_p,\end{aligned} \tag{S5}$$

which is the first claim. In particular $\bar{C}_{i,t}^{\mathsf{T}}\bar{R}_i^{-1}\bar{C}_{i,t}$ is invertible, so $K_{i,t}$ in (5) is well defined, and $K_{i,t}\bar{C}_{i,t} = I_p$.

Step 2 (decomposition, and properties of the noise and the bias). Applying $K_{i,t}$ to (S4) and using $K_{i,t}\bar{C}_{i,t} = I_p$ gives the decomposition (6), with $\eta_i(t) = K_{i,t}\bar{w}_{m,i,t}$ and $\beta_i(t) = K_{i,t}\bar{b}_{i,t}$. The exploration points are designed before the data are collected (Section 2 of the main paper), so the matrices $C_{i,t}$ and $K_{i,t}$ are deterministic; when C_i is constant, as in case study (A) of the main paper, this holds for arbitrary exploration points. Hence $\eta_i(t)$ is a deterministic linear function of the measurement noises $w_{m,i}(t),\dots,w_{m,i}(t+k-1)$: it is Gaussian and zero mean, and it is independent of the parameters $\theta_i(\cdot)$, of the process noises $w_i(\cdot)$, of the data of the other agents, and therefore of every $\beta_i(s)$, which by (S3) and (S4) is a linear function of $\theta_i(s), w_i(s),\dots,w_i(s+k-2)$. Since these, and the measurement noises, are independent across agents, $\eta_i(\cdot)$ and $\beta_i(\cdot)$ are also independent of $\eta_j(\cdot)$ and $\beta_j(\cdot)$ for every $j \neq i$. For $|t-s| \geq k$ the windows $\{t,\dots,t+k-1\}$ and $\{s,\dots,s+k-1\}$ are disjoint, so $\eta_i(t)$ and $\eta_i(s)$ are functions of disjoint sets of independent noises and are independent. Finally, $\theta_i(t)$ and the $w_i(\cdot)$ are zero mean, so $\mathbb{E}[\bar{b}_{i,t}] = 0$ and $\mathbb{E}[\beta_i(t)] = 0$. These are the properties stated after (6) in the main paper.

Step 3 (noise). By Step 2, $\eta_i(t)$ is zero mean with covariance

$$\begin{aligned}\Sigma_{\eta,i}(t) = K_{i,t}\bar{R}_i K_{i,t}^{\mathsf{T}} &= \big(\bar{C}_{i,t}^{\mathsf{T}}\bar{R}_i^{-1}\bar{C}_{i,t}\big)^{-1}\bar{C}_{i,t}^{\mathsf{T}}\bar{R}_i^{-1}\bar{C}_{i,t}\big(\bar{C}_{i,t}^{\mathsf{T}}\bar{R}_i^{-1}\bar{C}_{i,t}\big)^{-1} \\ &= \big(\bar{C}_{i,t}^{\mathsf{T}}\bar{R}_i^{-1}\bar{C}_{i,t}\big)^{-1} \preceq \alpha_{k,i}^{-1} I_p,\end{aligned} \tag{S6}$$

by (S5), which is the second claim; hence $\mathbb{E}\|\eta_i(t)\|^2 = \operatorname{tr}\Sigma_{\eta,i}(t) \le p/\alpha_{k,i}$. The gain is bounded as follows. Since $\bar{R}_i \succeq \lambda_{\min}(R_i) I_{kn}$,

$$\begin{aligned} K_{i,t}K_{i,t}^{\mathsf{T}} &\preceq \frac{K_{i,t}\bar{R}_i K_{i,t}^{\mathsf{T}}}{\lambda_{\min}(R_i)} \preceq \frac{I_p}{\alpha_{k,i}\lambda_{\min}(R_i)}, \qquad \text{so} \\ \|K_{i,t}\|^2 &\le \frac{\lambda_{\max}(R_i)}{\lfloor k/k_0 \rfloor\, \alpha_0\, \lambda_{\min}(R_i)} \le \frac{2k_0\lambda_{\max}(R_i)}{k\,\alpha_0\,\lambda_{\min}(R_i)} = \frac{\chi_i^2}{k}, \end{aligned} \tag{S7}$$

where the last inequality uses $\lfloor k/k_0 \rfloor \ge k/(2k_0)$, which holds because $k/k_0 \ge 1$.

Step 4 (bias). The two terms of (S3) are uncorrelated, since $\theta_i(t)$ is independent of the zero-mean noises $w_i(t), \dots, w_i(t+j-1)$. For the first term, the identity $A^j - I = (A-I)\sum_{l=0}^{j-1} A^l$ and $\|A^l\| \le C_\rho \rho_A^l \le C_\rho$ give $\|A^j - I\| \le C_\rho\, j\, \|A - I\|$, so $\mathbb{E}\|(A^j - I)\theta_i(t)\|^2 \le C_\rho^2 j^2 \|A-I\|^2 B_{\theta,i}^2$ by Step 0. For the second term, the summands are independent and zero mean, so

$$\mathbb{E}\Big\| \sum_{l=0}^{j-1} A^{j-1-l} w_i(t+l) \Big\|^2 = \sum_{l=0}^{j-1} \operatorname{tr}\left(A^{j-1-l} Q_i (A^{j-1-l})^{\mathsf{T}}\right) \le \sum_{l=0}^{j-1} \|A^{j-1-l}\|^2 \operatorname{tr} Q_i \le C_\rho^2\, j\, \operatorname{tr} Q_i.$$

Hence

$$\mathbb{E}\|\Delta_{i,t}(j)\|^2 \le C_\rho^2 \big(j^2 \|A - I\|^2 B_{\theta,i}^2 + j \operatorname{tr} Q_i \big), \qquad j = 0, \dots, k-1. \tag{S8}$$

With $\|C_{i,t+j}\| \le \Gamma_i$ (finite by the hypothesis of Lemma 1), $\sum_{j=1}^{k-1} j^2 = \frac{(k-1)k(2k-1)}{6} \le \frac{k^3}{3}$, and $\sum_{j=1}^{k-1} j = \frac{k(k-1)}{2} \le \frac{k^2}{2}$,

$$\mathbb{E}\|\bar{b}_{i,t}\|^2 = \sum_{j=1}^{k-1} \mathbb{E}\|C_{i,t+j}\Delta_{i,t}(j)\|^2 \le \Gamma_i^2 C_\rho^2 \Big(\frac{k^3}{3} \|A - I\|^2 B_{\theta,i}^2 + \frac{k^2}{2} \operatorname{tr} Q_i \Big). \tag{S9}$$

Combining (S7), (S9), and $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$,

$$\big(\mathbb{E}\|\beta_i(t)\|^2\big)^{1/2} \le \|K_{i,t}\| \big(\mathbb{E}\|\bar{b}_{i,t}\|^2\big)^{1/2} \le \frac{\chi_i}{\sqrt{k}} \Gamma_i C_\rho \Big(\frac{k^{3/2}}{\sqrt{3}} \|A - I\| B_{\theta,i} + \frac{k}{\sqrt{2}} \sqrt{\operatorname{tr} Q_i} \Big) = b_{k,i}. \tag{S10}$$

Step 5 (conclusion). By (6), Minkowski's inequality, (S6), and (S10),

$$\big(\mathbb{E}\|\tilde{\theta}_i(t) - \theta_i(t)\|^2\big)^{1/2} \le \big(\mathbb{E}\|\eta_i(t)\|^2\big)^{1/2} + \big(\mathbb{E}\|\beta_i(t)\|^2\big)^{1/2} \le \sqrt{p/\alpha_{k,i}} + b_{k,i} = \epsilon_i^{\mathrm{w}},$$

which is (14). Minkowski's inequality once more, with Step 0, gives the moment bound used in the later proofs,

$$\big(\mathbb{E}\|\tilde{\theta}_i(t)\|^2\big)^{1/2} \le \big(\mathbb{E}\|\theta_i(t)\|^2\big)^{1/2} + \epsilon_i^{\mathrm{w}} \le B_{\theta,i} + \epsilon_i^{\mathrm{w}} =: B'_{\theta,i}, \qquad t \in \{0, \dots, N-k\}. \tag{S11}$$

■

2. Concentration of the sample moments

Theorem 2 of the main paper involves a concentration constant c and a burn-in $N_0(\cdot)$ for which the bound (16) on the weighted sums of the sample moments (8) holds; the next lemma provides them. Throughout this section and the next, $\mathcal{T}_N := \{k, \dots, N-k-1\}$ is the index set of the sums in (8) and $N' := N - 2k = |\mathcal{T}_N|$. The sample moments are

quadratic forms in the Gaussian data of one agent, and the agents are independent; no mixing-time argument is needed, because the window estimates are linear in the Gaussian noise and their correlations decay geometrically in the time lag.

Lemma S1 (Concentration of the sample moments) *Let Assumptions 2 and 3 hold, $k \geq k_0$, and $\Gamma_i < \infty$ for every $i \in \mathcal{V}$. Let c_0 be the absolute constant of the Hanson–Wright inequality (Lemma S2), let $\bar{\chi}$, $\bar{\Gamma}$, $\bar{B}_\theta$ be the maxima over $i \in \mathcal{V}$ of χ_i, Γ_i, $B_{\theta,i}$, and let*

$$c_1 := \bar{\chi}^2 \bar{\Gamma}^2 C_\rho \bar{B}_\theta^2 + \frac{2k_0 \max_i \lambda_{\max}(R_i)}{k\,\alpha_0}, \qquad \kappa_{\mathrm{mix}} := 2k + \frac{2}{1-\rho_A}, \qquad c_2 := c_1\sqrt{\kappa_{\mathrm{mix}}}, \tag{S12}$$

$$c := p\,c_2\sqrt{\frac{2\big(1+\log_2(2p^2)\big)}{c_0}}, \qquad N_0(\delta') := 4k + \frac{\kappa_{\mathrm{mix}}}{c_0}\log\frac{2p^2}{\delta'}. \tag{S13}$$

Then, for every $j \in \{0,1\}$, every $\omega \in \mathbb{R}^m$, every $\delta' \in (0,1/2]$, and every $N \geq N_0(\delta')$, with probability at least $1-\delta'$,

$$\Big\| \sum_{l=1}^{m} \omega_l \big(\hat{M}_{j,l} - M_{j,l}\big) \Big\| \leq c\,\|\omega\| \sqrt{\frac{\log(1/\delta')}{N}}, \tag{S14}$$

which is (16).

Proof *Step 1 (Gaussian representation of the window estimates).* Fix $i \in \mathcal{V}$ and stack all the randomness of agent i during data collection in

$$\zeta_i := \mathrm{col}\,\big(\theta_i(0),\, w_i(0), \ldots, w_i(N-2),\, w_{m,i}(0), \ldots, w_{m,i}(N-1)\big) \in \mathbb{R}^d, \qquad d := N(p+n).$$

By (2) and the independence of its components, ζ_i is a zero-mean Gaussian vector with a block-diagonal covariance $\Sigma_{\zeta,i}$, and $\zeta_i = \Sigma_{\zeta,i}^{1/2} g_i$ with $g_i \sim \mathcal{N}(0, I_d)$. Iterating (2), every $\theta_i(t)$ with $t < N$ is a linear function of ζ_i, and so is every window estimate: by (5) and (S4),

$$\tilde{\theta}_i(t) = K_{i,t}\big(\Psi_{i,t}\Theta_{i,t} + \bar{w}_{m,i,t}\big), \qquad \Theta_{i,t} := \mathrm{col}\,\big(\theta_i(t), \ldots, \theta_i(t+k-1)\big), \tag{S15}$$

with $\Psi_{i,t} := \mathrm{blkdiag}(C_{i,t}, \ldots, C_{i,t+k-1})$, where $K_{i,t}$ and $\Psi_{i,t}$ are deterministic (Step 2 of Section 1). Hence $\tilde{\theta}_i(t) = L_{i,t} g_i$ for deterministic matrices $L_{i,t} \in \mathbb{R}^{p \times d}$, $t \in \{0, \ldots, N-k\}$: all window estimates of agent i are jointly Gaussian and zero mean.

Step 2 (geometric decay of the correlations). Let $G_i(t,s) := \mathbb{E}[\tilde{\theta}_i(t)\tilde{\theta}_i(s)^{\mathsf{T}}] = L_{i,t} L_{i,s}^{\mathsf{T}}$. Since $\eta_i(\cdot)$ is zero mean and independent of $\theta_i(\cdot)$ (Step 2 of Section 1), (S15) gives

$$G_i(t,s) = K_{i,t}\Psi_{i,t}\,\mathbb{E}\big[\Theta_{i,t}\Theta_{i,s}^{\mathsf{T}}\big]\,\Psi_{i,s}^{\mathsf{T}} K_{i,s}^{\mathsf{T}} + \mathbb{E}\big[\eta_i(t)\eta_i(s)^{\mathsf{T}}\big]. \tag{S16}$$

The (j,j') block of $\mathbb{E}[\Theta_{i,t}\Theta_{i,s}^{\mathsf{T}}]$ is $\mathbb{E}[\theta_i(t+j)\theta_i(s+j')^{\mathsf{T}}]$, which equals $A^{t+j-s-j'}P_i(s+j')$ if $t+j \geq s+j'$ and $P_i(t+j)(A^{s+j'-t-j})^{\mathsf{T}}$ otherwise, because the process noise after the earlier of the two times is zero mean and independent of the state at that time. Since $\|P_i(\cdot)\| \leq \mathrm{tr}\,P_i(\cdot) \leq B_{\theta,i}^2$ and $|j-j'| \leq k-1$,

$$\big\|\mathbb{E}[\theta_i(t+j)\theta_i(s+j')^{\mathsf{T}}]\big\| \leq C_\rho B_{\theta,i}^2 \rho_A^{|t-s+j-j'|} \leq C_\rho B_{\theta,i}^2 \rho_A^{(|t-s|-k+1)_+}.$$

A matrix with $k \times k$ blocks has norm at most k times the largest block norm, and $\|K_{i,t}\Psi_{i,t}\| \leq \|K_{i,t}\|\|\Psi_{i,t}\| \leq \chi_i\Gamma_i/\sqrt{k}$ by (S7), so the first term of (S16) has norm at most $\chi_i^2\Gamma_i^2 C_\rho B_{\theta,i}^2 \rho_A^{(|t-s|-k+1)_+}$. The second term vanishes for $|t-s| \geq k$ (Step 2 of Section 1) and otherwise, by Lemma S6 and (S6), has norm at most $\|\Sigma_{\eta,i}(t)\|^{1/2}\|\Sigma_{\eta,i}(s)\|^{1/2} \leq \alpha_{k,i}^{-1} \leq$

$2k_0\lambda_{\max}(R_i)/(k\alpha_0)$, where the last step is the bound $\lfloor k/k_0 \rfloor \geq k/(2k_0)$ of Step 3 of Section 1. Hence, with c_1 as in (S12),

$$\begin{aligned} &\|G_i(t,s)\| \leq c_1\, \rho_A^{(|t-s|-k+1)_+} \quad \text{for all } t,s, \qquad \text{and} \\ &\sum_{s\in\mathbb{Z}} \rho_A^{(|t-s|-k+1)_+} \leq (2k-1) + 2\sum_{j\geq 1} \rho_A^{j} \leq \kappa_{\mathrm{mix}}, \end{aligned} \tag{S17}$$

where the $2k-1$ integers s with $|t-s| \leq k-1$ contribute 1 each and the others contribute $\rho_A^{|t-s|-k+1}$.

Step 3 (the sample moments are Gaussian quadratic forms). Fix $j \in \{0,1\}$ and an entry (a,b), and let $u_t := L_{i,t+j}^{\mathsf{T}} e_a$ and $v_t := L_{i,t-k}^{\mathsf{T}} e_b$ for $t \in \mathcal{T}_N$, so that $e_a^{\mathsf{T}} \tilde{\theta}_i(t+j) = u_t^{\mathsf{T}} g_i$ and $e_b^{\mathsf{T}} z_i(t) = e_b^{\mathsf{T}} \tilde{\theta}_i(t-k) = v_t^{\mathsf{T}} g_i$. Then, by (8),

$$(\hat{M}_{j,i})_{ab} = \frac{1}{N'} \sum_{t\in\mathcal{T}_N} (u_t^{\mathsf{T}} g_i)(v_t^{\mathsf{T}} g_i) = g_i^{\mathsf{T}} B_i g_i, \qquad B_i := \frac{1}{2N'} \sum_{t\in\mathcal{T}_N} \big(u_t v_t^{\mathsf{T}} + v_t u_t^{\mathsf{T}}\big), \tag{S18}$$

with B_i symmetric, and $(M_{j,i})_{ab} = \mathbb{E}[g_i^{\mathsf{T}} B_i g_i]$. Every inner product between the vectors u_t, v_s is an entry of a correlation matrix: $u_t^{\mathsf{T}} u_s = e_a^{\mathsf{T}} G_i(t+j, s+j) e_a$, $v_t^{\mathsf{T}} v_s = e_b^{\mathsf{T}} G_i(t-k, s-k) e_b$, and $u_t^{\mathsf{T}} v_s = e_a^{\mathsf{T}} G_i(t+j, s-k) e_b$. By (S17),

$$|u_t^{\mathsf{T}} u_s| \leq c_1 \rho_A^{(|t-s|-k+1)_+}, \qquad |v_t^{\mathsf{T}} v_s| \leq c_1 \rho_A^{(|t-s|-k+1)_+}, \qquad |u_t^{\mathsf{T}} v_s| \leq c_1 \rho_A^{(|t-s+j+k|-k+1)_+}, \tag{S19}$$

and each of the three bounds, summed over $s \in \mathcal{T}_N$ for fixed t, or over t for fixed s, is at most $c_1\kappa_{\mathrm{mix}}$, by (S17) after the change of variable $s \mapsto s-j-k$ in the third case.

Step 4 (norms of B_i). Expanding $\|B_i\|_F^2 = \operatorname{tr}(B_i^2)$ with $\operatorname{tr}(u_t v_t^{\mathsf{T}} u_s v_s^{\mathsf{T}}) = (v_t^{\mathsf{T}} u_s)(v_s^{\mathsf{T}} u_t)$ and $\operatorname{tr}(u_t v_t^{\mathsf{T}} v_s u_s^{\mathsf{T}}) = (v_t^{\mathsf{T}} v_s)(u_s^{\mathsf{T}} u_t)$,

$$\begin{aligned} \|B_i\|_F^2 &= \frac{1}{2N'^2} \sum_{t,s\in\mathcal{T}_N} \Big[(u_t^{\mathsf{T}} u_s)(v_t^{\mathsf{T}} v_s) + (u_t^{\mathsf{T}} v_s)(u_s^{\mathsf{T}} v_t)\Big] \\ &\leq \frac{1}{2N'^2} \sum_{t\in\mathcal{T}_N} \Big[c_1 \sum_{s\in\mathcal{T}_N} |u_t^{\mathsf{T}} u_s| + c_1 \sum_{s\in\mathcal{T}_N} |u_t^{\mathsf{T}} v_s|\Big] \leq \frac{c_1^2 \kappa_{\mathrm{mix}}}{N'} = \frac{c_2^2}{N'}, \end{aligned} \tag{S20}$$

where we bounded $|v_t^{\mathsf{T}} v_s|$ and $|u_s^{\mathsf{T}} v_t|$ by c_1 and then used Step 3. For the operator norm, let $U := [u_t]_{t\in\mathcal{T}_N}$ and $V := [v_t]_{t\in\mathcal{T}_N}$ be the $d \times N'$ matrices with these columns, so that $B_i = \frac{1}{2N'}(UV^{\mathsf{T}} + VU^{\mathsf{T}})$ and $\|B_i\| \leq \|U\|\|V\|/N'$. The Gram matrix $U^{\mathsf{T}}U$ is symmetric positive semidefinite with entries $u_t^{\mathsf{T}} u_s$, so Gershgorin's theorem and Step 3 give $\|U\|^2 = \|U^{\mathsf{T}}U\| \leq \max_t \sum_s |u_t^{\mathsf{T}} u_s| \leq c_1\kappa_{\mathrm{mix}}$, and likewise $\|V\|^2 \leq c_1\kappa_{\mathrm{mix}}$. Hence

$$\|B_i\| \leq \frac{c_1\kappa_{\mathrm{mix}}}{N'}. \tag{S21}$$

Step 5 (Hanson–Wright inequality and union bound). The agents are independent, so $g := \operatorname{col}(g_1, \ldots, g_m) \sim \mathcal{N}(0, I_{md})$, and by (S18)

$$\sum_{l=1}^{m} \omega_l \big(\hat{M}_{j,l} - M_{j,l}\big)_{ab} = g^{\mathsf{T}} B_\omega g - \mathbb{E}\big[g^{\mathsf{T}} B_\omega g\big], \qquad B_\omega := \operatorname{blkdiag}\big(\omega_1 B_1, \ldots, \omega_m B_m\big),$$

where, by (S20) and (S21),

$$\|B_\omega\|_F^2 = \sum_{l=1}^{m} \omega_l^2 \|B_l\|_F^2 \leq \frac{\|\omega\|^2 c_2^2}{N'}, \qquad \|B_\omega\| = \max_l |\omega_l| \|B_l\| \leq \frac{\|\omega\|\, c_1\kappa_{\mathrm{mix}}}{N'}.$$

Let $\ell := \log(2p^2/\delta')/c_0$ and $s := \|\omega\| c_2 \sqrt{\ell/N'}$. Then $s^2/\|B_\omega\|_F^2 \geq \ell$, and, since $c_2/(c_1\kappa_{\mathrm{mix}}) = \kappa_{\mathrm{mix}}^{-1/2}$, $s/\|B_\omega\| \geq \sqrt{\ell N'/\kappa_{\mathrm{mix}}} \geq \ell$ provided that $N' \geq \kappa_{\mathrm{mix}}\ell$; so Lemma S2 gives

$$\mathbb{P}\Big(\Big|\sum_{l=1}^{m} \omega_l \big(\hat{M}_{j,l} - M_{j,l}\big)_{ab}\Big| > s\Big) \leq 2\exp(-c_0\ell) = \frac{\delta'}{p^2}.$$

A union bound over the p^2 entries shows that, with probability at least $1-\delta'$, every entry of $\sum_l \omega_l(\hat{M}_{j,l} - M_{j,l})$ has absolute value at most s, and then

$$\Big\|\sum_{l=1}^{m} \omega_l \big(\hat{M}_{j,l} - M_{j,l}\big)\Big\| \leq \Big\|\sum_{l=1}^{m} \omega_l \big(\hat{M}_{j,l} - M_{j,l}\big)\Big\|_F \leq p\, s = p\, c_2 \|\omega\| \sqrt{\frac{\log(2p^2/\delta')}{c_0 N'}}.$$

Step 6 (the constants). If $N \geq N_0(\delta')$ then $N \geq 4k$, so $N' = N - 2k \geq N/2$, and $N' \geq 2k + \kappa_{\mathrm{mix}}\ell \geq \kappa_{\mathrm{mix}}\ell$, so the proviso of Step 5 holds. Since $\delta' \leq 1/2$, $\log(1/\delta') \geq \log 2$ and

$$\log \frac{2p^2}{\delta'} = \log_2(2p^2)\, \log 2 + \log \frac{1}{\delta'} \leq \big(1 + \log_2(2p^2)\big) \log \frac{1}{\delta'}.$$

Inserting both bounds in the last display of Step 5 gives

$$p\, s \leq p\, c_2 \|\omega\| \sqrt{\frac{2\big(1 + \log_2(2p^2)\big)\log(1/\delta')}{c_0 N}} = c\, \|\omega\| \sqrt{\frac{\log(1/\delta')}{N}},$$

with c as in (S13), which is (S14). ∎

3. Proof of Theorem 2 (cooperative identification)

Proof Throughout, c and $N_0(\cdot)$ are the constant and the burn-in of Lemma S1, so that (16) holds; $\delta' := \delta/(4m) \leq 1/2$; $D_{j,l} := \hat{M}_{j,l} - M_{j,l}$; and $\mathcal{T}_N$, N' are as in Section 2.

Step 1 (what each agent holds after τ rounds). The recursion (9) is linear with the same weights for every entry of the moments, so $\hat{M}_j^{(i)}(\tau) = \sum_{l=1}^m (W^\tau)_{il} \hat{M}_{j,l}$. Since W^τ is doubly stochastic, $\sum_l (W^\tau)_{il} = 1$, and since $\sum_l (M_{j,l} - \bar{M}_j) = 0$,

$$\hat{M}_j^{(i)}(\tau) - \bar{M}_j = \underbrace{\frac{1}{m}\sum_{l=1}^{m} D_{j,l}}_{\text{pooled deviation}} + \underbrace{\sum_{l=1}^{m} \big((W^\tau)_{il} - \tfrac{1}{m}\big) D_{j,l}}_{\text{consensus deviation}} + \underbrace{\sum_{l=1}^{m} \big((W^\tau)_{il} - \tfrac{1}{m}\big)\big(M_{j,l} - \bar{M}_j\big)}_{\text{heterogeneity}}. \tag{S22}$$

The weights of the second and third terms form the vector $\omega^{(i)} := (W^\tau - J)^{\mathsf{T}} e_i$. Since W is doubly stochastic, $WJ = JW = J = J^2$, so $(W-J)^{r+1} = (W^r - J)(W - J) = W^{r+1} - J$ for every $r \geq 1$ with $(W-J)^1 = W - J$, that is, $W^\tau - J = (W-J)^\tau$. Hence $\|\omega^{(i)}\| \leq \|W^\tau - J\| = \|(W-J)^\tau\| \leq \|W - J\|^\tau = \sigma^\tau$.

Step 2 (the event $\mathcal{E}_\delta$). Apply Lemma S1 with confidence δ', which is admissible because $\delta' \leq 1/2$ and $N \geq N_0(\delta')$ by hypothesis, to the two pooled sums, $\omega = \frac{1}{m}\mathbf{1}_m$ with $\|\omega\| = m^{-1/2}$ and $j \in \{0,1\}$, and to the $2m$ consensus sums, $\omega = \omega^{(i)}$ with $i \in \mathcal{V}$ and $j \in \{0,1\}$. Let $\mathcal{E}_\delta$ be the intersection of these $2m + 2 \leq 4m$ events. Then $\mathbb{P}(\mathcal{E}_\delta) \geq 1 - (2m+2)\delta' \geq 1 - \delta$, and on $\mathcal{E}_\delta$ the first two terms of (S22) have norms at most $c\, m^{-1/2}\sqrt{\log(4m/\delta)/N}$ and $c\,\sigma^\tau \sqrt{\log(4m/\delta)/N}$, since $\log(1/\delta') = \log(4m/\delta)$. The third term has norm at most

$\|\omega^{(i)}\|_1 D_M \le \sqrt{m}\,\|\omega^{(i)}\| D_M \le \sqrt{m}\,\sigma^\tau D_M$, where $\|\cdot\|_1$ is the ℓ_1 norm. Hence, with $\epsilon^{\mathrm{m}}(N,\tau,\delta)$ as in (15), on $\mathcal{E}_\delta$,

$$\big\|\hat{M}_j^{(i)}(\tau) - \bar{M}_j\big\| \le \epsilon^{\mathrm{m}}(N,\tau,\delta) \qquad \text{for all } i \in \mathcal{V} \text{ and } j \in \{0,1\}. \tag{S23}$$

Step 3 (population moments). Fix $i \in \mathcal{V}$ and $t \in \mathcal{T}_N$. By (6), $\mathbb{E}[\tilde{\theta}_i(t)\tilde{\theta}_i(t-k)^\mathsf{T}]$ expands into nine terms. By Step 2 of Section 1, $\eta_i(t)$ is zero mean and independent of $\theta_i(t-k)$, of $\beta_i(t-k)$, and of $\eta_i(t-k)$; likewise, $\eta_i(t-k)$ is zero mean and independent of $\theta_i(t)$ and of $\beta_i(t)$. The five terms that contain a noise therefore vanish, and $\mathbb{E}[\theta_i(t)\theta_i(t-k)^\mathsf{T}] = A^k P_i(t-k)$ by (2). Averaging over $t \in \mathcal{T}_N$ identifies the term $\Delta_{0,i}$ of the main paper:

$$\begin{aligned} M_{0,i} &= A^k \Sigma_{\theta,i} + \Delta_{0,i}, \\ \Delta_{0,i} &= A^k\big(\bar{P}_{k,i} - \Sigma_{\theta,i}\big) + \frac{1}{N'}\sum_{t\in\mathcal{T}_N} \mathbb{E}\Big[\theta_i(t)\beta_i(t-k)^\mathsf{T} + \beta_i(t)\theta_i(t-k)^\mathsf{T} + \beta_i(t)\beta_i(t-k)^\mathsf{T}\Big], \end{aligned} \tag{S24}$$

with $\bar{P}_{k,i} := \frac{1}{N'}\sum_{t\in\mathcal{T}_N} P_i(t-k)$. The first part of $\Delta_{0,i}$ is the transient of the initial covariance: by (S2), $\|\bar{P}_{k,i} - \Sigma_{\theta,i}\| \le \frac{1}{N'}\sum_{s\ge 0} C_\rho^2 \rho_A^{2s} \|P_{i,0} - \Sigma_{\theta,i}\|$, and $\|A^k\| \le C_\rho$; it vanishes when $P_{i,0} = \Sigma_{\theta,i}$. The second part collects the terms involving the windowing bias; it is bounded by Lemma S6 with $(\mathbb{E}\|\theta_i(\cdot)\|^2)^{1/2} \le B_{\theta,i}$ and $(\mathbb{E}\|\beta_i(\cdot)\|^2)^{1/2} \le b_{k,i}$, the latter by (S10). Altogether,

$$\|\Delta_{0,i}\| \le \frac{C_\rho^3 \|P_{i,0} - \Sigma_{\theta,i}\|}{N'(1-\rho_A^2)} + b_{k,i}\big(2B_{\theta,i} + b_{k,i}\big). \tag{S25}$$

Next, multiply (7) from the right by $z_i(t)^\mathsf{T} = \tilde{\theta}_i(t-k)^\mathsf{T}$ and take expectations. The instrument $z_i(t)$ is a function of $\theta_i(t-k)$, of $w_i(t-k),\dots,w_i(t-2)$, and of $w_{m,i}(t-k),\dots,w_{m,i}(t-1)$. The process noise $w_i(t)$ and the noise residual $\eta_i(t+1) - A\eta_i(t)$, a function of $w_{m,i}(t),\dots,w_{m,i}(t+k)$, are zero mean and independent of it, so they drop out:

$$M_{1,i} = AM_{0,i} + \Xi_i, \qquad \Xi_i := \frac{1}{N'}\sum_{t\in\mathcal{T}_N} \mathbb{E}\Big[\big(\beta_i(t+1) - A\beta_i(t)\big) z_i(t)^\mathsf{T}\Big], \tag{S26}$$

which is the relation stated in Section 3 of the main paper, and $\|\Xi_i\| \le (1+\|A\|)\, b_{k,i}\, B'_{\theta,i}$ by Lemma S6, (S10), and $(\mathbb{E}\|z_i(t)\|^2)^{1/2} \le B'_{\theta,i}$, which is (S11). Averaging (S24) and (S26) over the agents,

$$\bar{M}_0 = A^k\bar{\Sigma}_\theta + \bar{\Delta}_0, \qquad \bar{M}_1 = A\bar{M}_0 + \bar{\Xi}, \qquad \|\bar{\Xi}\| \le \epsilon^{\mathrm{b}} := \frac{1+\|A\|}{m}\sum_{i=1}^m b_{k,i} B'_{\theta,i}, \tag{S27}$$

with $\bar{\Xi} := \frac{1}{m}\sum_i \Xi_i$. This is the sense in which the moments are additive across agents: every $M_{1,i}$ equals A times $M_{0,i}$ up to the bias residual Ξ_i, with the same A for all agents, so the pooled moments satisfy the same relation.

Step 4 (invertibility). Weyl's inequality for singular values, $\sigma_{\min}(X+Y) \ge \sigma_{\min}(X) - \|Y\|$, the hypothesis $\|\bar{\Delta}_0\| \le \nu_k/2$, (S23), and the hypothesis $\epsilon^{\mathrm{m}} \le \nu_k/8$ give

$$\sigma_{\min}(\bar{M}_0) \ge \sigma_{\min}(A^k\bar{\Sigma}_\theta) - \|\bar{\Delta}_0\| \ge \frac{\nu_k}{2}, \qquad \sigma_{\min}\big(\hat{M}_0^{(i)}(\tau)\big) \ge \sigma_{\min}(\bar{M}_0) - \epsilon^{\mathrm{m}} \ge \frac{3\nu_k}{8} \quad \text{on } \mathcal{E}_\delta, \tag{S28}$$

so that $\bar{M}_0$ and, on $\mathcal{E}_\delta$, every $\hat{M}_0^{(i)}(\tau)$ are invertible, with $\|\bar{M}_0^{-1}\| \le 2/\nu_k$ and $\|(\hat{M}_0^{(i)}(\tau))^{-1}\| \le 8/(3\nu_k)$. (Here $\nu_k > 0$ because A is invertible and $\bar{\Sigma}_\theta \succ 0$ by Step 0 of Section 1.)

Step 5 (perturbation of the estimator). Work on $\mathcal{E}_\delta$, fix i, and omit the argument τ and the superscript (i). By (S27), $\hat{M}_1\bar{M}_0^{-1} = A + \bar{\Xi}\bar{M}_0^{-1}$, and the identity $X^{-1} - Y^{-1} = -Y^{-1}(X-Y)X^{-1}$ with $X = \hat{M}_0$ and $Y = \bar{M}_0$ gives

$$\hat{A}^{(i)} - A = \hat{M}_1\hat{M}_0^{-1} - \bar{M}_1\bar{M}_0^{-1} + \bar{\Xi}\bar{M}_0^{-1} = \big(\hat{M}_1 - \bar{M}_1\big)\hat{M}_0^{-1} - \bar{M}_1\bar{M}_0^{-1}\big(\hat{M}_0 - \bar{M}_0\big)\hat{M}_0^{-1} + \bar{\Xi}\bar{M}_0^{-1}. \tag{S29}$$

By (S23), (S28), and $\|\bar{M}_1\bar{M}_0^{-1}\| \le \|A\| + 2\epsilon^{\mathrm{b}}/\nu_k$, the three terms have norms at most $\frac{8}{3\nu_k}\epsilon^{\mathrm{m}}$, $\big(\|A\| + \frac{2\epsilon^{\mathrm{b}}}{\nu_k}\big)\frac{8}{3\nu_k}\epsilon^{\mathrm{m}}$, and $\frac{2\epsilon^{\mathrm{b}}}{\nu_k}$, so that

$$\|\hat{A}^{(i)} - A\| \le \frac{8}{3\nu_k}\Big(1 + \|A\| + \frac{2\epsilon^{\mathrm{b}}}{\nu_k}\Big)\epsilon^{\mathrm{m}}(N,\tau,\delta) + \frac{2\epsilon^{\mathrm{b}}}{\nu_k} \qquad \text{on } \mathcal{E}_\delta, \text{ for all } i \in \mathcal{V}. \tag{S30}$$

Step 6 (the constants). Insert (15) into (S30) and use $m^{-1/2} + \sigma^\tau = (1 + \sqrt{m}\sigma^\tau)/\sqrt{m}$. The first term of (S30) becomes the finite-sample and heterogeneity terms of (17) with

$$\begin{aligned} C_A &:= \frac{8c}{3\nu_k}\Big(1 + \|A\| + \frac{2\epsilon^{\mathrm{b}}}{\nu_k}\Big), \qquad C_W := \frac{C_A}{c}, \\ C_B &:= \frac{2(1+\|A\|)C_\rho}{\nu_k}\max_i\big(\Gamma_i\chi_i B'_{\theta,i}\big)\max\Big\{\frac{\bar{B}_\theta}{\sqrt{3}}, \frac{1}{\sqrt{2}}\Big\}, \end{aligned} \tag{S31}$$

where $\bar{B}_\theta := \max_i B_{\theta,i}$. For the second term of (S30), (S1) and Jensen's inequality $\frac{1}{m}\sum_i \sqrt{\operatorname{tr} Q_i} \le \sqrt{\frac{1}{m}\sum_i \operatorname{tr} Q_i} = \sqrt{\operatorname{tr}\bar{Q}}$, with $\bar{Q} := \frac{1}{m}\sum_i Q_i$, give

$$\begin{aligned} \frac{2\epsilon^{\mathrm{b}}}{\nu_k} &= \frac{2(1+\|A\|)}{\nu_k}\,\frac{1}{m}\sum_{i=1}^m b_{k,i}B'_{\theta,i} \\ &\le \frac{2(1+\|A\|)C_\rho}{\nu_k}\max_i\big(\Gamma_i\chi_i B'_{\theta,i}\big)\Big(\frac{k\|A-I\|\bar{B}_\theta}{\sqrt{3}} + \sqrt{\frac{k}{2}\operatorname{tr}\bar{Q}}\Big) \\ &\le C_B\Big(k\|A-I\| + \sqrt{k\operatorname{tr}\bar{Q}}\Big), \end{aligned}$$

which is the bias-residual term and completes the proof of (17). The constants C_A, C_W, C_B depend on k and on the problem data, but not on N, τ, or δ. ∎

4. Proof of Lemma 3 (gradient tracking with a moving minimizer)

Proof Let $X(t)$, $S(t)$, $G(t) \in \mathbb{R}^{m\times n}$ stack the iterates $x_i(t)$, the trackers $s_i(t)$, and the gradient signals $\gamma_i(t) = \nabla_x \hat{f}_i(x_i(t), t)$, $t \ge N$, row-wise, and let $\bar{s}(t) := \frac{1}{m}\sum_i s_i(t)$, $\bar{g}(t) := \frac{1}{m}\sum_i \gamma_i(t)$, and $\hat{\theta}(t) := \operatorname{col}(\hat{\theta}_1(t), \dots, \hat{\theta}_m(t))$. Since the Euclidean norm of a stacked vector is the Frobenius norm of the corresponding stack, the errors of (18) read

$$\begin{aligned} a_t &= \|X(t) - \mathbf{1}_m\bar{x}(t)^{\mathsf{T}}\|_F, \qquad b_t = \sqrt{m}\,\|\bar{x}(t) - \hat{x}^*(t)\|, \\ c_t &= \|S(t) - \mathbf{1}_m\bar{s}(t)^{\mathsf{T}}\|_F, \qquad d_t = \|\hat{\theta}(t+1) - \hat{\theta}(t)\|, \end{aligned} \tag{S32}$$

and $e_t = \operatorname{col}(a_t, b_t, c_t)$. For $t \ge N$ the forecasts are fixed functions of the data, so (11) is a deterministic recursion in the iterates and the trackers, and every statement below holds for every realization of the data. By hypothesis, $\hat{\theta}_i(t) \in \Theta_i$ for every i, that is, $\hat{\theta}(t) \in \Theta = \Theta_1 \times \cdots \times \Theta_m$, for all $t \ge N$, so Assumption 1 applies to the forecast costs: for every $t \ge N$, $\hat{F}(\cdot, t) := \sum_i \hat{f}_i(\cdot, t)$ is μ-strongly convex and mL-smooth (its Hessian is the

sum of the m Hessians of the $\hat{f}_i(\cdot,t)$, each of norm at most L), each $\hat{f}_i(\cdot,t)$ is L-smooth, and $\hat{x}^*(t)$ is the unique minimizer of $\hat{F}(\cdot,t)$, so that $\nabla_x \hat{F}(\hat{x}^*(t),t)=0$. In matrix form, (11) reads

$$X(t+1)=W\big(X(t)-\alpha S(t)\big),\qquad S(t+1)=W\big(S(t)+G(t+1)-G(t)\big). \tag{S33}$$

We use $\mathbf{1}_m^\mathsf{T}W=\mathbf{1}_m^\mathsf{T}$, $JW=J$ (hence $(I-J)W=W-J$), $(W-J)\mathbf{1}_m=0$, $\|W-J\|=\sigma$, $\|W\|\le 1$, and $\|W-I\|\le 2$, all consequences of Assumption 4: W is symmetric with eigenvalues in $[-1,1]$, and, being irreducible with a positive diagonal, it is primitive, so 1 is a simple eigenvalue, -1 is not an eigenvalue, and σ, the largest modulus of the other eigenvalues, is smaller than 1.

Step 0 (tracking identity). Multiplying (S33) by $\frac{1}{m}\mathbf{1}_m^\mathsf{T}$ gives $\bar{x}(t+1)=\bar{x}(t)-\alpha\bar{s}(t)$ and $\bar{s}(t+1)=\bar{s}(t)+\bar{g}(t+1)-\bar{g}(t)$ for all $t\ge 0$ (with the oracle gradients in $\bar{g}(t)$ for $t<N$). Since $\bar{s}(0)=\bar{g}(0)$, induction gives $\bar{s}(t)=\bar{g}(t)$ for all $t\ge 0$; in particular,

$$\bar{s}(t)=\bar{g}(t)=\frac{1}{m}\sum_{i=1}^m \nabla_x \hat{f}_i(x_i(t),t)\qquad\text{and}\qquad \bar{x}(t+1)=\bar{x}(t)-\alpha\bar{g}(t)\qquad\text{for all } t\ge N. \tag{S34}$$

Step 1 (disagreement of the iterates). By (S33), $(I-J)W=W-J$, and $(W-J)\mathbf{1}_m=0$,

$$X(t+1)-\mathbf{1}_m\bar{x}(t+1)^\mathsf{T}=(I-J)X(t+1)=(W-J)\Big[\big(X(t)-\mathbf{1}_m\bar{x}(t)^\mathsf{T}\big)-\alpha\big(S(t)-\mathbf{1}_m\bar{s}(t)^\mathsf{T}\big)\Big], \tag{S35}$$

so that $a_{t+1}\le \sigma\,(a_t+\alpha c_t)$.

Step 2 (optimization error of the average). Write $\bar{g}(t)=\frac{1}{m}\nabla_x\hat{F}(\bar{x}(t),t)+r_t$ with

$$r_t:=\frac{1}{m}\sum_{i=1}^m\Big(\nabla_x\hat{f}_i(x_i(t),t)-\nabla_x\hat{f}_i(\bar{x}(t),t)\Big),\qquad \|r_t\|\le\frac{L}{m}\sum_{i=1}^m\|x_i(t)-\bar{x}(t)\|\le\frac{L}{\sqrt{m}}\,a_t, \tag{S36}$$

by the L-smoothness of $\hat{f}_i(\cdot,t)$ and by $\sum_i\|x_i-\bar{x}\|\le\sqrt{m}\,(\sum_i\|x_i-\bar{x}\|^2)^{1/2}$ (Cauchy–Schwarz inequality). The map $T_t(x):=x-\frac{\alpha}{m}\nabla_x\hat{F}(x,t)$ is a gradient step on $\hat{F}(\cdot,t)/m$, which is (μ/m)-strongly convex and L-smooth, and $\hat{x}^*(t)$ is its fixed point. Its Jacobian $I-\frac{\alpha}{m}\nabla_x^2\hat{F}(x,t)$ is symmetric with eigenvalues in $[1-\alpha L,\,1-\alpha\mu/m]\subseteq[0,\,1-\alpha\mu/m]$, because $\alpha L\le(1-\sigma)^2/4\le 1$ by (19); hence, by the mean value theorem in integral form, $T_t(x)-T_t(y)=\int_0^1\big(I-\frac{\alpha}{m}\nabla_x^2\hat{F}(y+\varsigma(x-y),t)\big)\,d\varsigma\,(x-y)$, T_t is a contraction with modulus $1-\alpha\mu/m$. By (S34), $\bar{x}(t+1)=T_t(\bar{x}(t))-\alpha r_t$, and by (4) (Lemma S5, which also gives $L_*<\infty$), applicable since $\hat{\theta}(t),\hat{\theta}(t+1)\in\Theta$, the predicted minimizer moves by at most $\|\hat{x}^*(t+1)-\hat{x}^*(t)\|\le L_*\|\hat{\theta}(t+1)-\hat{\theta}(t)\|=L_*d_t$. Therefore

$$\begin{aligned}\|\bar{x}(t+1)-\hat{x}^*(t+1)\|&\le\|T_t(\bar{x}(t))-T_t(\hat{x}^*(t))\|+\alpha\|r_t\|+\|\hat{x}^*(t)-\hat{x}^*(t+1)\|\\&\le\big(1-\tfrac{\alpha\mu}{m}\big)\|\bar{x}(t)-\hat{x}^*(t)\|+\frac{\alpha L}{\sqrt{m}}\,a_t+L_*d_t.\end{aligned} \tag{S37}$$

Multiplying (S37) by $\sqrt{m}$,

$$b_{t+1}\le\alpha L\,a_t+\big(1-\tfrac{\alpha\mu}{m}\big)b_t+\sqrt{m}\,L_*\,d_t. \tag{S38}$$

Step 3 (disagreement of the trackers). As in Step 1,

$$S(t+1)-\mathbf{1}_m\bar{s}(t+1)^\mathsf{T}=(W-J)\big(S(t)-\mathbf{1}_m\bar{s}(t)^\mathsf{T}\big)+(W-J)\big(G(t+1)-G(t)\big), \tag{S39}$$

so that $c_{t+1}\le\sigma\big(c_t+\|G(t+1)-G(t)\|_F\big)$. For each agent, split the change of the gradient signal into the change of the iterate and the change of the forecast, using $\nabla_x\hat{f}_i(x,t+1)-\nabla_x\hat{f}_i(x,t)=$

$C_i(x)\big(\hat{\theta}_i(t+1)-\hat{\theta}_i(t)\big)$ from (10); the L-smoothness of $\hat{f}_i(\cdot,t+1)$ and $\|C_i(x)\|\le\Gamma_i\le\Gamma$ then give

$$\begin{aligned}\|\gamma_i(t+1)-\gamma_i(t)\| &\le \big\|\nabla_x\hat{f}_i(x_i(t+1),t+1)-\nabla_x\hat{f}_i(x_i(t),t+1)\big\| \\ &\quad+\big\|C_i(x_i(t))\big(\hat{\theta}_i(t+1)-\hat{\theta}_i(t)\big)\big\| \\ &\le L\|x_i(t+1)-x_i(t)\|+\Gamma\|\hat{\theta}_i(t+1)-\hat{\theta}_i(t)\|. \end{aligned}\tag{S40}$$

Taking the Euclidean norm over the agents and using Minkowski's inequality, $\|G(t+1)-G(t)\|_F\le L\|X(t+1)-X(t)\|_F+\Gamma d_t$. By (S33) and $(W-I)\mathbf{1}_m=0$,

$$\begin{aligned}X(t+1)-X(t)&=(W-I)\big(X(t)-\mathbf{1}_m\bar{x}(t)^{\mathsf{T}}\big)-\alpha WS(t),\\ \|X(t+1)-X(t)\|_F&\le 2a_t+\alpha\|S(t)\|_F,\end{aligned}\tag{S41}$$

and $\|S(t)\|_F\le\|S(t)-\mathbf{1}_m\bar{s}(t)^{\mathsf{T}}\|_F+\|\mathbf{1}_m\bar{s}(t)^{\mathsf{T}}\|_F=c_t+\sqrt{m}\|\bar{s}(t)\|$, where, by (S34), (S36), $\nabla_x\hat{F}(\hat{x}^*(t),t)=0$, and the mL-smoothness of $\hat{F}(\cdot,t)$,

$$\|\bar{s}(t)\|\le\frac{1}{m}\|\nabla_x\hat{F}(\bar{x}(t),t)-\nabla_x\hat{F}(\hat{x}^*(t),t)\|+\|r_t\|\le L\|\bar{x}(t)-\hat{x}^*(t)\|+\frac{L}{\sqrt{m}}\,a_t=\frac{L}{\sqrt{m}}(a_t+b_t).\tag{S42}$$

Collecting (S39)–(S42),

$$c_{t+1}\le\sigma\Big(L(2+\alpha L)\,a_t+\alpha L^2\,b_t+(1+\alpha L)\,c_t+\Gamma\,d_t\Big).\tag{S43}$$

Step 4 (the error recursion and its spectral radius). Inequalities (S35), (S38), and (S43) read, entrywise, $e_{t+1}\le\Lambda e_t+v\,d_t$ for all $t\ge N$, with

$$\Lambda:=\begin{bmatrix}\sigma & 0 & \alpha\sigma\\ \alpha L & 1-\alpha\mu/m & 0\\ \sigma L(2+\alpha L) & \sigma\alpha L^2 & \sigma(1+\alpha L)\end{bmatrix},\qquad v:=\begin{bmatrix}0\\ \sqrt{m}\,L_*\\ \sigma\Gamma\end{bmatrix}.\tag{S44}$$

The entries of Λ and v are nonnegative: by (19), $\alpha\mu/m\le\mu/\big(m(\kappa+1)L\big)=\mu^2/\big(m(mL+\mu)L\big)<1$. We show that $\rho(\Lambda)<1$ with Lemma S3. The matrix

$$D:=I-\Lambda=\begin{bmatrix}1-\sigma & 0 & -\alpha\sigma\\ -\alpha L & \alpha\mu/m & 0\\ -\sigma L(2+\alpha L) & -\sigma\alpha L^2 & 1-\sigma(1+\alpha L)\end{bmatrix}$$

has nonpositive off-diagonal entries, and expanding its determinant along the first row gives the leading principal minors

$$\begin{aligned}D_1&=1-\sigma,\qquad D_2=(1-\sigma)\frac{\alpha\mu}{m},\\ \det D&=\alpha\Big[\frac{\mu}{m}(1-\sigma)\big(1-\sigma-\sigma\alpha L\big)-\sigma^2\alpha^2L^3-\sigma^2\alpha L\frac{\mu}{m}(2+\alpha L)\Big].\end{aligned}\tag{S45}$$

The first two are positive because $\sigma<1$. For the third, let $\epsilon:=1-\sigma\in(0,1]$ and $x:=\alpha L$. The step-size condition (19) gives $x\le\epsilon^2/4\le 1/4$ and $\alpha L^2\le L/(\kappa+1)=L\mu/(mL+\mu)<\mu/m$, so that $\sigma^2\alpha^2L^3=\sigma^2x\,(\alpha L^2)\le\sigma^2x\,\mu/m$ and

$$\begin{aligned}\frac{m}{\alpha\mu}\det D&\ge\epsilon(\epsilon-\sigma x)-\sigma^2x-\sigma^2x(2+x)=\epsilon^2-x\big(\sigma\epsilon+\sigma^2(3+x)\big)\\ &\ge\epsilon^2-\frac{\epsilon^2}{4}\Big(\sigma\epsilon+\frac{13}{4}\sigma^2\Big)\ge\epsilon^2\Big(1-\frac{1}{4}\Big(\frac{1}{4}+\frac{13}{4}\Big)\Big)=\frac{\epsilon^2}{8},\end{aligned}$$

where the second inequality uses $x \le \epsilon^2/4$ and $x \le 1/4$, and the third uses $\sigma\epsilon = \sigma(1-\sigma) \le 1/4$ and $\sigma^2 \le 1$. Hence

$$\det D \ge \frac{\alpha\mu}{8m}\,(1-\sigma)^2 > 0, \tag{S46}$$

and Lemma S3 gives $\rho(\Lambda) < 1$.

Step 5 (conclusion). Since Λ and v are entrywise nonnegative, the recursion $e_{t+1} \le \Lambda e_t + v\, d_t$ can be unrolled from $t = N$:

$$0 \le e_t \le \Lambda^{t-N} e_N + \sum_{q=N}^{t-1} \Lambda^{t-1-q}\, v\, d_q \qquad \text{entrywise, for all } t \ge N. \tag{S47}$$

Fix $\rho_{\mathrm{gt}} \in (\rho(\Lambda), 1)$ and let $c_\Lambda := \sup_{j\ge 0} \|\Lambda^j\|/\rho_{\mathrm{gt}}^j$, which is finite by Gelfand's formula $\lim_j \|\Lambda^j\|^{1/j} = \rho(\Lambda) < \rho_{\mathrm{gt}}$. Taking norms in (S47) (the Euclidean norm is monotone in the entries of a nonnegative vector),

$$\|e_t\| \le c_\Lambda \Big(\rho_{\mathrm{gt}}^{t-N} \|e_N\| + \|v\| \sum_{q=N}^{t-1} \rho_{\mathrm{gt}}^{t-1-q} d_q \Big),$$

and finally $\|x_i(t) - \hat{x}^*(t)\| \le \|x_i(t) - \bar{x}(t)\| + \|\bar{x}(t) - \hat{x}^*(t)\| \le a_t + b_t/\sqrt{m} \le \sqrt{2}\,\|e_t\|$, which gives (20) with $c_{\mathrm{gt}} := \sqrt{2}\, c_\Lambda \max\{1, \|v\|\}$. Both ρ_{gt} and c_{gt} depend only on Λ and v, that is, on $(\sigma, \mu, L, m, \alpha, \Gamma, L_*)$. ■

5. Proof of Theorem 4 (distributed tracking error)

Proof Throughout, $t \ge N$ and $H = t - (N-k) \ge k \ge 1$, so that $\hat{\theta}_i(t) = (\hat{A}^{(i)})^H \tilde{\theta}_i(N-k)$ by (10); $\mathbb{E}_\delta[Z] = \mathbb{E}[Z\mathbb{I}_{\mathcal{E}_\delta}]$ for the event $\mathcal{E}_\delta$ of Theorem 2; and $B'_{\theta,i} = B_{\theta,i} + \epsilon_i^{\mathrm{w}}$ is the moment bound (S11).

Step 1 (powers of the identified dynamics). On $\mathcal{E}_\delta$, Theorem 2 gives $\|\hat{A}^{(i)} - A\| \le \epsilon^{\mathrm{id}}(\delta)$ for every i, so Lemma S4 with $E = \hat{A}^{(i)} - A$ and $\epsilon = \epsilon^{\mathrm{id}}(\delta)$ yields

$$\|(\hat{A}^{(i)})^j\| \le C_\rho \hat{\rho}^j \quad \text{for all } j \ge 0, \qquad \rho(\hat{A}^{(i)}) \le \hat{\rho} < 1 \qquad \text{on } \mathcal{E}_\delta. \tag{S48}$$

The telescoping identity $(\hat{A}^{(i)})^H - A^H = \sum_{l=0}^{H-1} (\hat{A}^{(i)})^l (\hat{A}^{(i)} - A) A^{H-1-l}$ and $\rho_A \le \hat{\rho}$ then give, on $\mathcal{E}_\delta$,

$$\|(\hat{A}^{(i)})^H - A^H\| \le \sum_{l=0}^{H-1} C_\rho \hat{\rho}^{\,l}\, \epsilon^{\mathrm{id}}(\delta)\, C_\rho \rho_A^{H-1-l} \le C_\rho^2 H \hat{\rho}^{\,H-1} \epsilon^{\mathrm{id}}(\delta). \tag{S49}$$

Step 2 (forecast error: bound (22)). By (13), $\hat{\theta}_i(t) - \theta_i(t)$ is the sum of three terms. On $\mathcal{E}_\delta$ the first is bounded through (S49); dropping the indicator and using Jensen's inequality $\mathbb{E}\|Z\| \le (\mathbb{E}\|Z\|^2)^{1/2}$ together with (S11) and (14), and writing $S_{H,i} := \sum_{j=0}^{H-1} A^j w_i(t-1-j)$

for the third term,

$$\mathbb{E}_\delta\big\|\big((\hat{A}^{(i)})^H - A^H\big)\tilde{\theta}_i(N-k)\big\| \le C_\rho^2 H \hat{\rho}^{\,H-1}\epsilon^{\mathrm{id}}(\delta)\,\mathbb{E}\|\tilde{\theta}_i(N-k)\| \le C_\rho^2 H \hat{\rho}^{\,H-1}\epsilon^{\mathrm{id}}(\delta)\,B'_{\theta,i}, \tag{S50}$$

$$\mathbb{E}_\delta\|A^H\tilde{e}_i\| \le \|A^H\|\,\mathbb{E}\|\tilde{e}_i\| \le C_\rho \rho_A^H\,\epsilon_i^{\mathrm{w}}, \tag{S51}$$

$$\mathbb{E}_\delta\|S_{H,i}\| \le \big(\mathbb{E}\|S_{H,i}\|^2\big)^{1/2} = \Big(\operatorname{tr}\sum_{j=0}^{H-1} A^j Q_i (A^j)^{\mathsf{T}}\Big)^{1/2} = \sigma_{H,i}, \tag{S52}$$

where (S52) uses that the summands of $S_{H,i}$ are independent and zero mean; $\sigma_{H,i}^2 \le \operatorname{tr}\Sigma_{\theta,i}$ because the series defining $\Sigma_{\theta,i}$ (Step 0 of Section 1) has positive semidefinite terms. Summing (S50)–(S52) and using the triangle inequality in (13) proves $\mathbb{E}_\delta\|\hat{\theta}_i(t) - \theta_i(t)\| \le \epsilon_i^{\mathrm{f}}(N,t;\delta)$, which is (22).

Step 3 (prediction error). Since $\theta(t) \in \Theta$ (Assumption 1) and $\hat{\theta}(t) \in \Theta$ (hypothesis of Lemma 3), (4) (Lemma S5) and the bound $\|\mathrm{col}(v_1,\dots,v_m)\| \le \sum_j \|v_j\|$ give

$$\|\hat{x}^*(t) - x^*(t)\| \le L_*\|\hat{\theta}(t) - \theta(t)\| \le L_* \sum_{j=1}^m \|\hat{\theta}_j(t) - \theta_j(t)\|, \tag{S53}$$

and multiplying by $\mathbb{I}_{\mathcal{E}_\delta}$, taking expectations, and using Step 2 gives term (iii) of (21).

Step 4 (tracking of the predicted minimizer). Lemma 3 holds for every realization of the data, with the forecasts of that realization; its hypotheses are among those of the theorem. The drift of the forecasts is bounded, on $\mathcal{E}_\delta$, using (10), (S48), and $\|\hat{A}^{(j)} - I\| \le \|A - I\| + \|\hat{A}^{(j)} - A\| \le \|A - I\| + \epsilon^{\mathrm{id}}(\delta)$: for $q \ge N$,

$$\begin{aligned} d_q &\le \sum_{j=1}^m \|\hat{\theta}_j(q+1) - \hat{\theta}_j(q)\| = \sum_{j=1}^m \big\|(\hat{A}^{(j)})^{q-(N-k)}\big(\hat{A}^{(j)} - I\big)\tilde{\theta}_j(N-k)\big\| \\ &\le C_\rho \hat{\rho}^{\,q-N+k}\big(\|A - I\| + \epsilon^{\mathrm{id}}(\delta)\big)\sum_{j=1}^m \|\tilde{\theta}_j(N-k)\|. \end{aligned} \tag{S54}$$

With $\rho_\star = \max\{\rho_{\mathrm{gt}}, \hat{\rho}\}$, the weighted sum of the drifts in (20) is bounded by a geometric series,

$$\sum_{q=N}^{t-1} \rho_{\mathrm{gt}}^{\,t-1-q}\hat{\rho}^{\,q-N+k} \le \hat{\rho}^{\,k}\sum_{q=N}^{t-1}\rho_\star^{\,t-1-q}\rho_\star^{\,q-N} = \hat{\rho}^{\,k}(t-N)\rho_\star^{\,t-N-1}, \tag{S55}$$

so Lemma 3, (S54), and (S55) give, on $\mathcal{E}_\delta$,

$$\|x_i(t) - \hat{x}^*(t)\| \le c_{\mathrm{gt}}\rho_{\mathrm{gt}}^{\,t-N}\|e_N\| + c_{\mathrm{gt}}C_\rho\hat{\rho}^{\,k}(t-N)\rho_\star^{\,t-N-1}\big(\|A - I\| + \epsilon^{\mathrm{id}}(\delta)\big)\sum_{j=1}^m\|\tilde{\theta}_j(N-k)\|. \tag{S56}$$

Multiplying by $\mathbb{I}_{\mathcal{E}_\delta}$, taking expectations, and using $\mathbb{E}\|\tilde{\theta}_j(N-k)\| \le B'_{\theta,j}$, by Jensen's inequality and (S11), yields terms (i) and (ii) of (21), with $U_N = (\|A - I\| + \epsilon^{\mathrm{id}}(\delta))\sum_j B'_{\theta,j}$.

Step 5 (finiteness of $\mathbb{E}_\delta\|e_N\|$). Fix any $\theta^0 \in \Theta$, which is nonempty by Assumption 1. For every x, the L-smoothness of $f_i(\cdot,\theta_i^0)$ and $\|C_i(x)\| \le \Gamma$ give $\|\nabla_x f_i^0(x)\| \le \|\nabla_x f_i(x,\theta_i^0)\| + \|C_i(x)\theta_i^0\| \le \|\nabla_x f_i(0,\theta_i^0)\| + L\|x\| + \Gamma\|\theta_i^0\|$, so $\nabla_x f_i^0$ grows at most linearly. For $t < N$ the gradient signal is the oracle output, with $\|\gamma_i(t)\| \le \|\nabla_x f_i^0(x_i(t))\| + \Gamma\|\theta_i(t)\| + \|w_{m,i}(t)\|$,

and for $t = N$ it is the forecast gradient, with $\|\gamma_i(N)\| \le \|\nabla_x f_i^0(x_i(N))\| + \Gamma\|\hat{\theta}_i(N)\|$. Since $x_i(t+1)$ and $s_i(t+1)$ in (11) are linear in the $x_j(t)$, $s_j(t)$, $\gamma_j(t)$, and $\gamma_j(t+1)$, induction over $t \in \{0, \dots, N\}$ shows that every $\|x_i(t)\|$ and $\|s_i(t)\|$ is bounded by an affine function of the norms of the Gaussian data and of $\|\hat{\theta}_j(N)\|$, $j \in \mathcal{V}$; and on $\mathcal{E}_\delta$, $\|\hat{\theta}_j(N)\| \le C_\rho \hat{\rho}^k \|\tilde{\theta}_j(N-k)\|$ by (S48), where $\tilde{\theta}_j(N-k)$ is Gaussian. Hence $\mathbb{E}_\delta a_N < \infty$ and $\mathbb{E}_\delta c_N < \infty$. For b_N, (4) with $\hat{\theta}(N), \theta^0 \in \Theta$ gives

$$b_N \le \sqrt{m}\big(\|\bar{x}(N)\| + \|x^*(\theta^0)\| + L_*\|\hat{\theta}(N) - \theta^0\|\big),$$

whose $\mathbb{E}_\delta$-expectation is finite for the same reasons. Therefore $\mathbb{E}_\delta\|e_N\| \le \mathbb{E}_\delta(a_N + b_N + c_N) < \infty$.

Step 6 (assembling). By (12), $\|x_i(t) - x^*(t)\| \le \|x_i(t) - \hat{x}^*(t)\| + \|\hat{x}^*(t) - x^*(t)\|$. Multiplying by $\mathbb{I}_{\mathcal{E}_\delta}$, taking expectations, and combining Steps 3 and 4 gives (21). ∎

References

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Appendix A. Auxiliary results

The following results are used in the proofs. The first two are quoted from the literature; the others are proved for completeness.

Lemma S2 (Hanson–Wright inequality) *There is an absolute constant $c_0 > 0$ such that for every $g \sim \mathcal{N}(0, I_D)$, every symmetric $B \in \mathbb{R}^{D\times D}$, and every $s \ge 0$ (Rudelson and Vershynin, 2013, Theorem 1.1),*

$$\mathbb{P}\big(|g^\mathsf{T} B g - \mathbb{E}[g^\mathsf{T} B g]| > s\big) \le 2\exp\Big(-c_0 \min\Big\{\frac{s^2}{\|B\|_F^2}, \frac{s}{\|B\|}\Big\}\Big).$$

Lemma S3 (M-matrix criterion) *Let $\Lambda \in \mathbb{R}^{q\times q}$ be entrywise nonnegative. If $I - \Lambda$ has positive leading principal minors, then $I - \Lambda$ is a nonsingular M-matrix and $\rho(\Lambda) < 1$ (Berman and Plemmons, 1994, Theorem 6.2.3).*

Lemma S4 (Powers of a perturbed matrix) *If $\|A^j\| \le C_\rho \rho_A^j$ for all $j \ge 0$, with $C_\rho \ge 1$, and $\|E\| \le \epsilon$, then $\|(A+E)^j\| \le C_\rho(\rho_A + C_\rho\epsilon)^j$ for all $j \ge 0$, and hence $\rho(A+E) \le \rho_A + C_\rho\epsilon$.*

Proof Expanding the product, $(A+E)^j$ is the sum of the 2^j words of length j in the letters A and E. A word with l letters E has the form $A^{a_0}EA^{a_1}E\cdots EA^{a_l}$ with $a_0 + \cdots + a_l = j - l$,

and there are $\binom{j}{l}$ such words. Since $C_\rho \geq 1$, the bound $\|A^a\| \leq C_\rho \rho_A^a$ holds also for $a = 0$, so each such word has norm at most $\prod_{r=0}^{l} \|A^{a_r}\| \cdot \|E\|^l \leq C_\rho^{l+1} \rho_A^{j-l} \epsilon^l$. Hence

$$\|(A+E)^j\| \leq \sum_{l=0}^{j} \binom{j}{l} C_\rho^{l+1} \rho_A^{j-l} \epsilon^l = C_\rho \sum_{l=0}^{j} \binom{j}{l} \rho_A^{j-l} (C_\rho \epsilon)^l = C_\rho (\rho_A + C_\rho \epsilon)^j,$$

and the bound on the spectral radius follows from Gelfand's formula $\rho(A+E) = \lim_j \|(A+E)^j\|^{1/j}$. ∎

Lemma S5 (Lipschitz continuity of the minimizer) *Let Assumption 1 hold. Then, for all $\theta, \theta' \in \Theta$, $\|x^*(\theta') - x^*(\theta)\| \leq L_* \|\theta' - \theta\|$ with $L_* = \mu^{-1} \sup_{\theta \in \Theta} \|C(x^*(\theta))\|$ and $C(x) = [C_1(x) \ \cdots \ C_m(x)]$; this is (4). If, moreover, $\Gamma_i = \sup_{x \in \mathbb{R}^n} \|C_i(x)\| < \infty$ for every $i \in \mathcal{V}$, then $L_* \leq \mu^{-1} \big(\sum_{i=1}^{m} \Gamma_i^2 \big)^{1/2}$.*

Proof By (1), $\nabla_x F(x,\theta) = \sum_i \nabla_x f_i^0(x) + \sum_i C_i(x)\theta_i = \sum_i \nabla_x f_i^0(x) + C(x)\theta$, so $\nabla_x F(x,\theta) - \nabla_x F(x,\theta') = C(x)(\theta - \theta')$ for every x. Let $x := x^*(\theta)$ and $x' := x^*(\theta')$; the optimality conditions read $\nabla_x F(x,\theta) = 0$ and $\nabla_x F(x',\theta') = 0$. By the μ-strong convexity of $F(\cdot,\theta)$, its gradient is μ-strongly monotone, so

$$\begin{aligned} \mu \|x' - x\|^2 &\leq \big\langle \nabla_x F(x',\theta) - \nabla_x F(x,\theta),\, x' - x \big\rangle = \big\langle \nabla_x F(x',\theta) - \nabla_x F(x',\theta'),\, x' - x \big\rangle \\ &= \big\langle C(x')(\theta - \theta'),\, x' - x \big\rangle \leq \|C(x')\| \|\theta' - \theta\| \|x' - x\|. \end{aligned}$$

Dividing by $\mu \|x' - x\|$ (if $x' \neq x$) gives $\|x' - x\| \leq \mu^{-1} \|C(x^*(\theta'))\| \|\theta' - \theta\| \leq L_* \|\theta' - \theta\|$. Finally, $\|C(x)\|^2 = \|\sum_i C_i(x) C_i(x)^\mathsf{T}\| \leq \sum_i \|C_i(x)\|^2 \leq \sum_i \Gamma_i^2$. ∎

Lemma S6 (Cross-covariances) *For random vectors $x \in \mathbb{R}^a$ and $y \in \mathbb{R}^b$ with finite second moments,*

$$\|\mathbb{E}[xy^\mathsf{T}]\| \leq \|\mathbb{E}[xx^\mathsf{T}]\|^{1/2} \|\mathbb{E}[yy^\mathsf{T}]\|^{1/2} \leq \big(\mathbb{E}\|x\|^2\big)^{1/2} \big(\mathbb{E}\|y\|^2\big)^{1/2}.$$

Proof For unit vectors $u \in \mathbb{R}^a$ and $v \in \mathbb{R}^b$, the Cauchy–Schwarz inequality gives

$$\begin{aligned} u^\mathsf{T} \mathbb{E}[xy^\mathsf{T}] v = \mathbb{E}[(u^\mathsf{T} x)(v^\mathsf{T} y)] &\leq \big(\mathbb{E}(u^\mathsf{T} x)^2\big)^{1/2} \big(\mathbb{E}(v^\mathsf{T} y)^2\big)^{1/2} \\ &= \big(u^\mathsf{T} \mathbb{E}[xx^\mathsf{T}] u\big)^{1/2} \big(v^\mathsf{T} \mathbb{E}[yy^\mathsf{T}] v\big)^{1/2} \leq \|\mathbb{E}[xx^\mathsf{T}]\|^{1/2} \|\mathbb{E}[yy^\mathsf{T}]\|^{1/2}; \end{aligned}$$

taking the supremum over u and v gives the first inequality. The second follows from $\|\mathbb{E}[xx^\mathsf{T}]\| \leq \operatorname{tr} \mathbb{E}[xx^\mathsf{T}] = \mathbb{E}\|x\|^2$, since $\mathbb{E}[xx^\mathsf{T}]$ is positive semidefinite. ∎